

The Neuron as an Electric Cable

An Electromagnetic Simulation Project

$$C_m \frac{dV}{dt} = I_{\text{ext}} - \bar{g}_{Na} m^3 h (V - E_{Na}) - \bar{g}_K n^4 (V - E_K) - g_L (V - E_L)$$

The neuron membrane modelled as an active electrical circuit.

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Abstract. The goal of this project was to show that a neuron can be modelled using basic electromagnetics. The cell membrane behaves like a capacitor, the axon behaves like a cable with resistance and capacitance, and the electrical signal neurons use to communicate can be simulated with a set of differential equations. All simulations are done in Python and the results are shown below.

1 Introduction

The nervous system runs on electricity. Neurons communicate by sending voltage pulses — called *action potentials* — along their axons. The interesting thing is that the physics behind this is not special biology; it is the same electromagnetics we study in circuits and field theory.

The cell membrane is a thin insulating layer between two conductive fluids, so it acts as a capacitor. The axon is a long tube of conductive fluid with a leaky membrane wall, so it acts like a lossy cable. And the voltage obeys a differential equation that comes directly from applying KCL to a small segment of the cable.

This project simulates three things:

1. The electric field and capacitance of the cell membrane
2. An action potential at a single point using the Hodgkin–Huxley model
3. The action potential propagating along the axon

2 The Membrane as a Capacitor

The cell membrane is about 5 nm thick and made of lipids, which are insulators. The inside and outside of the cell are both filled with conductive ionic fluid. This gives the structure:

$$\underbrace{\text{Conductive fluid}}_{\text{outside}} - \underbrace{\text{Lipid membrane}}_{\text{insulator, } d \approx 5 \text{ nm}} - \underbrace{\text{Conductive fluid}}_{\text{inside}}$$

which is exactly a parallel-plate capacitor. The standard formulas apply:

$$E \approx \frac{V_m}{d} \quad C_m = \frac{\epsilon}{d} \quad w_e = \frac{1}{2} \epsilon E^2$$

At resting potential ($V_m \approx -70$ mV, $d = 5$ nm):

$$E = \frac{70 \times 10^{-3}}{5 \times 10^{-9}} = 1.4 \times 10^7 \text{ V/m}$$

That is 14 million volts per metre — a very strong field produced by a very small voltage, only because the membrane is so thin.

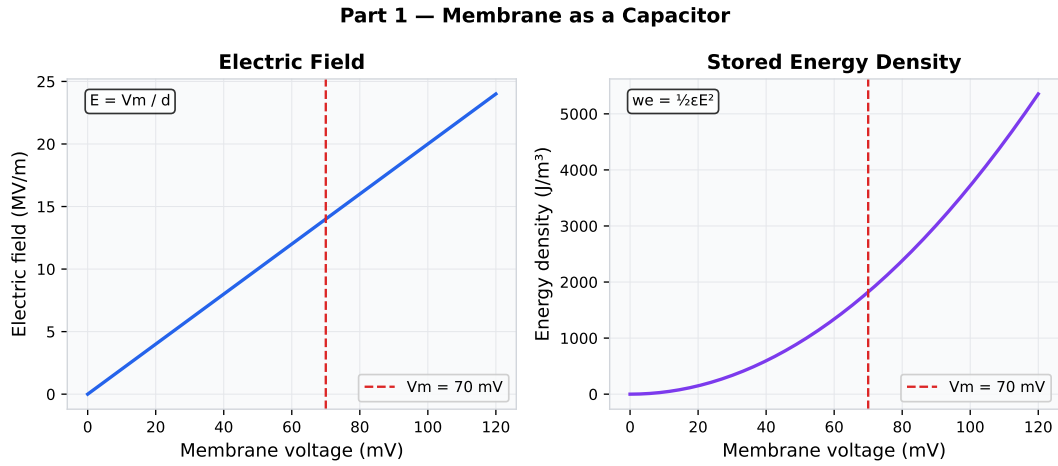


Figure 1. Left: electric field inside the membrane as a function of membrane voltage. Right: electrostatic energy density stored in the membrane. Both increase with voltage according to the parallel-plate capacitor model.

The specific capacitance (per unit area) is $C_m = \epsilon / d$. For the lipid membrane with $\epsilon_r \approx 2.1$ and $d = 5$ nm this gives roughly $0.37 \mu\text{F}/\text{m}^2$, which matches measured values for real cell membranes.

3 The Action Potential

3.1 Why a Passive Cable Is Not Enough

A simple RC cable loses voltage as the signal travels. To carry a signal over long distances without it decaying, neurons use *voltage-gated ion channels* — protein pores in the membrane that open or close depending on the local voltage.

When the voltage rises above a threshold, sodium (Na^+) channels open and ions rush in, raising the voltage further. This positive feedback produces a sharp spike. Potassium (K^+) channels then open more slowly and pull the voltage back down. The result is an *action potential*: a self-regenerating pulse that travels along the axon without decaying.

3.2 The Hodgkin–Huxley Model

Hodgkin and Huxley [?] modelled this with KCL at the membrane. The four currents are an external stimulus and three ionic currents:

$$C_m \frac{dV}{dt} = I_{\text{ext}} - \underbrace{\bar{g}_{\text{Na}} m^3 h (V - E_{\text{Na}})}_{I_{\text{Na}}} - \underbrace{\bar{g}_{\text{K}} n^4 (V - E_{\text{K}})}_{I_{\text{K}}} - \underbrace{g_{\text{L}} (V - E_{\text{L}})}_{I_{\text{L}}}$$

m, h, n are gating variables (0 to 1) saying how open each channel is. Each one follows a simple first-order equation:

$$\frac{dm}{dt} = \alpha_m(V) (1 - m) - \beta_m(V) m$$

The rate functions α and β depend on voltage, which is what makes the system nonlinear and capable of generating the spike.

Table 1. Hodgkin–Huxley model parameters used in the simulation.

Parameter	Symbol	Value	Meaning
Membrane capacitance	C_m	$1 \mu\text{F}/\text{cm}^2$	Charge storage
Max Na^+ conductance	$\bar{g}_{\text{Na}}$	$120 \text{ mS}/\text{cm}^2$	Na^+ channel strength
Max K^+ conductance	$\bar{g}_{\text{K}}$	$36 \text{ mS}/\text{cm}^2$	K^+ channel strength
Na^+ reversal pot.	E_{Na}	$+50 \text{ mV}$	Na^+ equilibrium
K^+ reversal pot.	E_{K}	-77 mV	K^+ equilibrium

Part 2 — Hodgkin-Huxley Action Potential

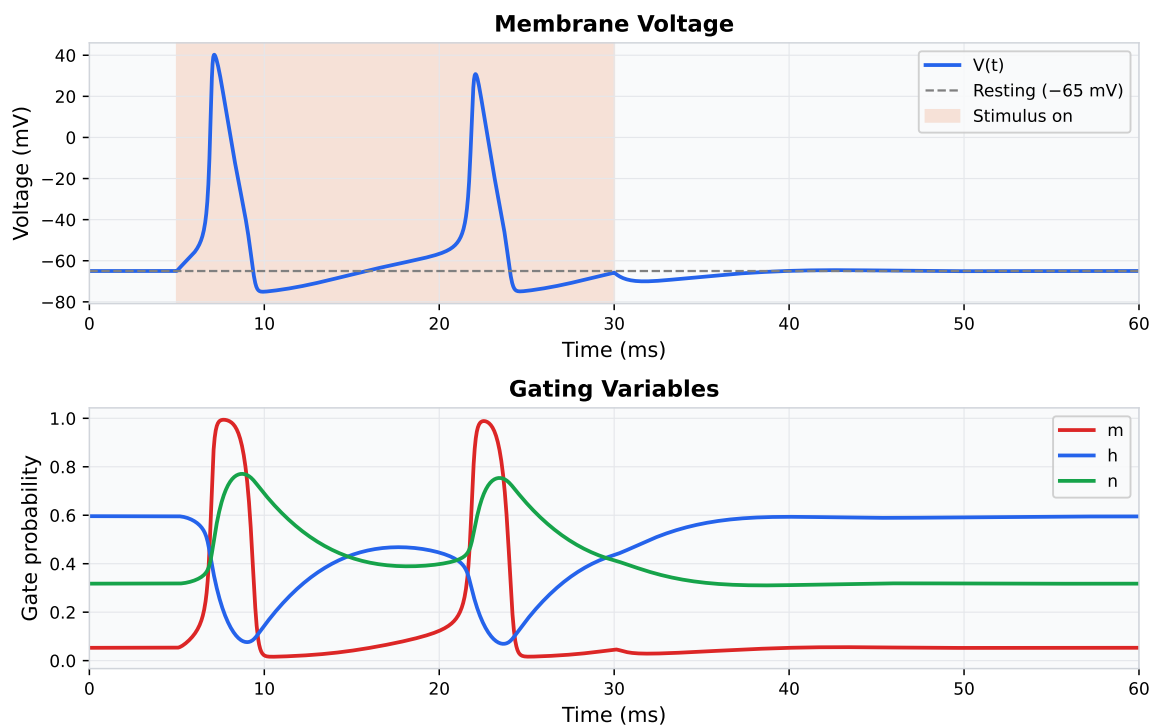


Figure 2. Hodgkin–Huxley simulation at a single point on the membrane. Top: membrane voltage — action potentials fire repeatedly while the stimulus current is applied. Bottom: the three gating variables m , h , n showing how the ion channels open and close during each spike.

In Figure 2, m rises sharply at the start of each spike (Na^+ channels opening) and h drops shortly after (Na^+ channels inactivating), while n rises more slowly (K^+ channels pulling the voltage back down). The spike fires again once the channels have reset.

4 Propagation Along the Axon

4.1 The Cable Model

The axon is divided into compartments. Each compartment has its own HH membrane with capacitance and ion channels, and adjacent compartments are coupled through

the axial resistance of the intracellular fluid. The axial resistance per unit length is:

$$r_i = \frac{1}{\sigma_i \pi a^2}$$

where a is the axon radius and σ_i is the conductivity of the axoplasm. The voltage in each compartment obeys:

$$C_m \frac{\partial V}{\partial t} = \frac{a}{2R_i} \frac{\partial^2 V}{\partial x^2} + I_{HH}(V, m, h, n)$$

where I_{HH} is the sum of all HH currents. This is the cable equation: the telegrapher's equation with $L' = 0$ and a nonlinear source term from the ion channels.

4.2 Simulation Result

A brief current pulse was injected at the left end of a 60 mm model axon (radius $50 \mu\text{m}$), and the voltage was tracked across all compartments over time.

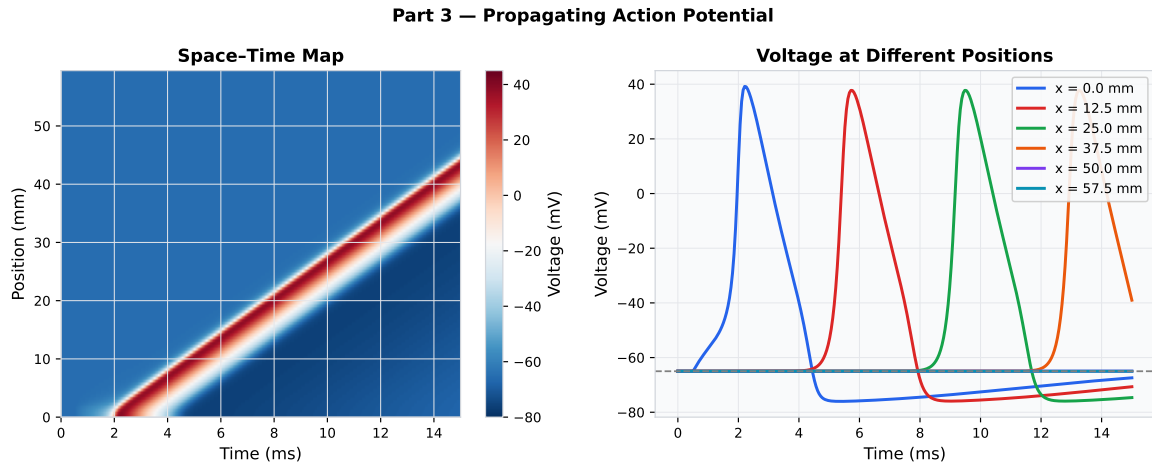


Figure 3. Action potential propagating along the axon. Left: space-time heatmap — the red band is the action potential traveling from left to right at roughly constant speed. Right: voltage traces at six positions show the same spike arriving later at each successive point.

Unlike a passive cable where the signal would decay, here the spike keeps its shape and amplitude along the full length. The ion channels at each compartment regenerate the signal locally, so the pulse is self-sustaining.

Conduction speed scales roughly as $\theta \propto \sqrt{a}$: a thicker axon has lower r_i , the voltage spreads further ahead, and the next compartment is triggered sooner. Myelinated axons (wrapped in an insulating sheath) increase the effective membrane resistance, which increases the space constant and allows even faster propagation.

5 Conclusion

The project shows that the electromagnetic tools we study in this course apply directly to a biological system. The membrane is a capacitor, the axon is a lossy cable, and the

action potential is a self-regenerating wave that travels along the cable without any loss.

The simulation reproduces real neuron behaviour: the action potential fires above threshold (Figure 2), travels at constant speed (Figure 3), and the membrane fields match the capacitor model (Figure 1).